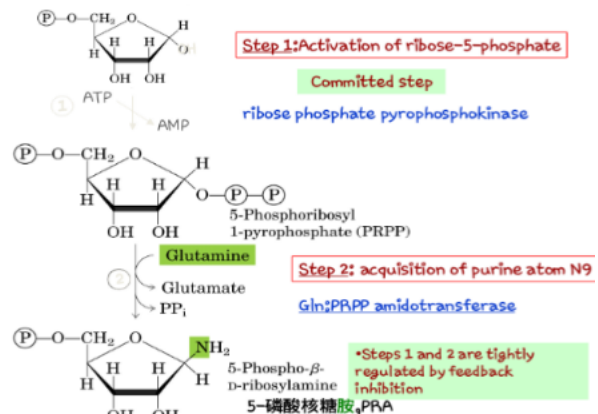


AAU 2K11/19 Module 3 Questions

(Includes remembered & repeated questions from previous years exams)

1. The rate limiting enzyme during de novo synthesis of purine nucleotides is

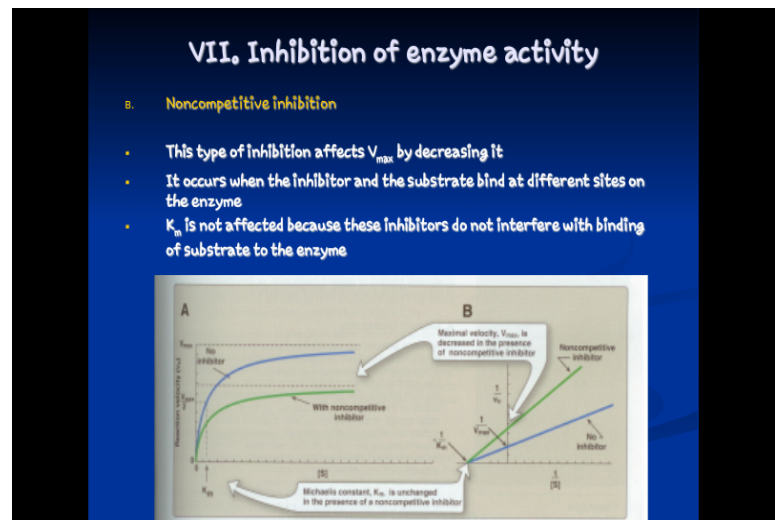
- A. PRPP Synthetase / Ribose phosphate pyrophosphokinase/
- B. Gln:PRPPamidotransferase
- C. Glycinamide synthetase
- D. GAR transformylase
- E. AIR carboxylase



ANS. A,.....PRPP Synthetase / Ribose phosphate pyrophosphokinase/ is the rate limiting enzyme catalyzing the first reaction, the committed step, during de novo synthesis of purine nucleotides

2. A non-competitive inhibitor

- A. Decreases V_{max}
- B. Increases V_{max}
- C. Increases K_m , doesn't affect V_{max}
- D. Increases K_m , increases V_{max}
- E. doesn't affect K_m , increases V_{max}



ANS. A,.....A non-competitive decreases V_{max} but doesn't affect K_m . A competitive inhibitor increases K_m but doesn't affect V_{max} .

3. The link reaction is catalyzed by

- A. Pyruvate kinase
- B. Pyruvate dehydrogenase
- C. Pyruvate carboxylase
- D.....
- E.....

ANS. B,.. (Echi tyake final lay yekerech nat 120gnawa, 119 tefetenin)

4. An endergonic reaction:

- A. Overall releases energy
- B. Overall Requires energy
- C.....
- D.....
- E...

ANS. B,..... Endergonic reactions require energy, Exergonic reactions release energy.

5. Which of the ff enzyme of gluconeogenesis is needed to bypass a reaction of glycolysis?

- A...
- B. Pyruvate kinase
- C....
- D. Pyruvate dehydrogenase
- E. Pyruvate carboxylase

Peculiar enzymes of gluconeogenesis

- In addition to seven enzymes of glycolysis, this pathway uses unique enzymes:
 - pyruvate carboxylase
 - Phosphoenolpyruvate carboxykinase
 - Fructose 1,6-bisphosphatase
 - Glucose 6- phosphatase
- Note the 1. two enzymes are required to bypass the Pyruvate kinase reaction, the 3. to bypass the PFK reaction and the 4. to bypass the Hexokinase reaction. All are one way reactions of glycolysis hence going back using the same enzymes is impossible.

ANS. E

6. Digestion of limit dextrins to disaccharides is catalyzed by

A. Pancreatic amylase

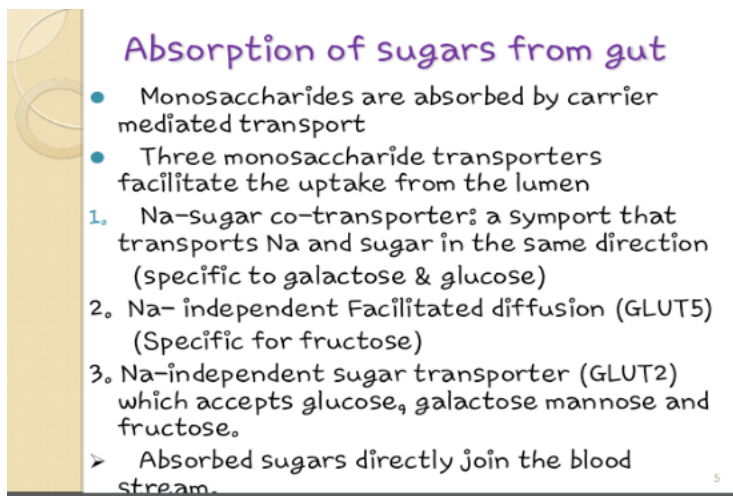
B. Oligosaccharidase.

C. Disaccharidases

D.....

E.....

ANS B,.....Gn if it was digestion of sugars into limit dextrins and disaccharides....melsu pancreatic amylase yehon neber



Absorption of sugars from gut

- Monosaccharides are absorbed by carrier mediated transport
- Three monosaccharide transporters facilitate the uptake from the lumen
 1. Na-sugar co-transporter: a symport that transports Na and sugar in the same direction (specific to galactose & glucose)
 2. Na-independent Facilitated diffusion (GLUT5) (Specific for fructose)
 3. Na-independent sugar transporter (GLUT2) which accepts glucose, galactose mannose and fructose.
- Absorbed sugars directly join the blood stream.

7. Fructose is absorbed from the gut by

A. Na-Sugar co-transporter

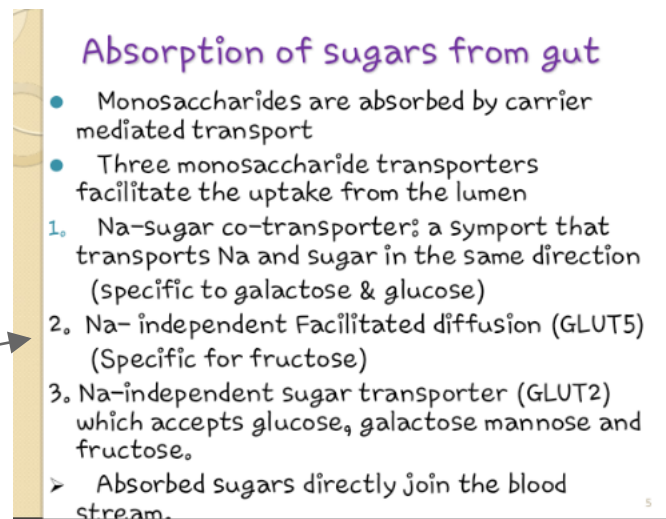
B. Na-independent sugar transport

C. Na-independent facilitated diffusion.

D.....

E.....

ANS. C



Absorption of sugars from gut

- Monosaccharides are absorbed by carrier mediated transport
- Three monosaccharide transporters facilitate the uptake from the lumen
 1. Na-sugar co-transporter: a symport that transports Na and sugar in the same direction (specific to galactose & glucose)
 2. Na-independent Facilitated diffusion (GLUT5) (Specific for fructose)
 3. Na-independent sugar transporter (GLUT2) which accepts glucose, galactose mannose and fructose.
- Absorbed sugars directly join the blood stream.

8. HMP(Pentose phosphate) pathway will take place in a high rate during..

A...

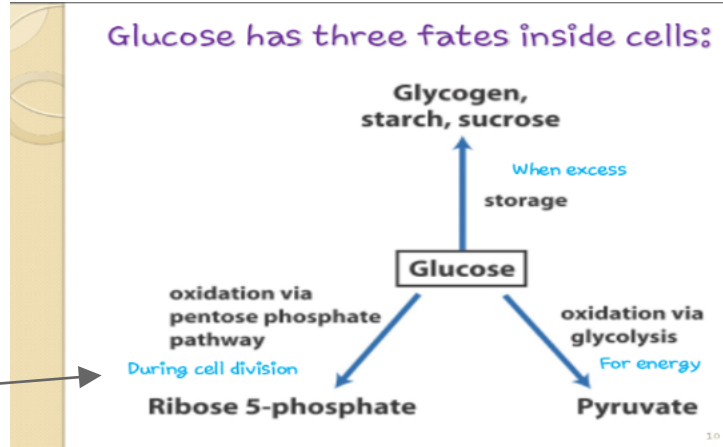
B...

C. Cell division.

D...

E...

ANS. C



9. $\text{RH}_2\text{OH} + \text{O}_2 \rightarrow \text{RHO} + \text{H}_2\text{O}$..

This reaction is catalyzed by the enzyme...

A...

B. Oxidase

C. Oxygenase

D.....

E.....

ANS. B,.....in the reaction 2 hydrogens are transferred to oxygen

10. In the above reaction (from question number 9), which cofactor/coenzyme is required as an electron acceptor?

A. Flavin adenine dinucleotide (FAD)

B.....

C.....

D.....

E.....

ANS. A

Oxygenases and oxidases

- Oxygenases catalyse the incorporation of oxygen into a substrate. dioxygenases incorporate both oxygen atoms into the substrate and mono oxygenases incorporate a single oxygen atom as a hydroxyl group into the substrate and the second oxygen atom is reduced to water.
- Oxidases catalyze the transfer of two electrons from a donor to oxygen, producing hydrogen peroxide. Usually a flavin coenzyme or metal cofactor is required. Refer to the following reaction. Oxidases catalyze this type of reaction.



11. The glucose transporter found in erythrocytes and brain.

- A. GLUT1 B. GLUT2 C. GLUT3 D. GLUT4 E. GLUT5

ANS. A

12. Which of the ff uses a glucose transporter that is insulin dependent

- A. Adipose/Fat D. Kidney
B. Brain E. Liver
C. Erythrocyte

ANS. A

13. Which of the ff uses a glucose transporter that is insulin dependent

- A. Erythrocyte D. Kidney
B. Brain E. Liver
C. Muscle

ANS. C

14. Which glucose transporter is most likely related to disease,...such as Diabetes mellitus

- A. GLUT1 B. GLUT2 C. GLUT3 D. GLUT4 E. GLUT5

ANS. D

Glucose Transporters on cells

- GLUT1- in erythrocytes and brain
- GLUT2- liver, kidney and intestine
- GLUT3- brain
- GLUT4- fat and muscle (insulin dependent)
- GLUT5 and GLUT7 distributed over many tissues

15. Which is true about hexokinase and glucokinase? (SEE Q. 114)

- A. Glucokinase is present in all tissues
- B. Glucokinase has broad substrate specificity than Hexokinase
- C. Hexokinase activity is increased by insulin
- D. Hexokinase has low K_m and more affinity for glucose than Glucokinase
- E. Glucokinase has low K_m and more affinity for glucose than Hexokinase

Hexokinase and Glucokinase

- Hexokinase is present in all tissues.
- Has broad substrate specificity, phosphorylates all hexoses (Glucose, mannose, Fructose)
- Hexokinase has low K_m hence high affinity for glucose
- Glucokinase is found in liver parenchymal cells and islets of Langerhans in pancreas
- Glucokinase has high K_m , so functions only when there is high glucose level in cells.
- Glucokinase is regulated by F-6-P and glucose
- Glucokinase activity is increased by insulin

17

ANS. D

16. Rat poison arsenate can inhibit which of the ff glycolytic enzymes..

- A. Pyruvate kinase
- B. Glyceraldehyde 3-P dehydrogenase
- C. Hexokinase
- D. PFK-1
- E.....

ANS. B,..... Toxicity of arsenic is explained primarily by inhibition of enzymes like PYRUVATE DEHYDROGENASE and GLYCERALDEHYDE 3-P DEHYDROGENASE, which need lipoic acid as cofactor

Arsenate Poisoning

- Toxicity of arsenic is explained primarily by inhibition of enzymes like pyruvate dehydrogenase and Glyceraldehyde 3-P dehydrogenase, which need lipoic acid as cofactor
- Pentavalent arsenic (arsenate) prevents ATP and NADH production from glycolysis without actually inhibiting the pathway
- Competes with P_i as a substrate for glyceraldehyde 3-P dehydrogenase, forming a complex that spontaneously hydrolyses to form 3-phosphoglycerate
- Bypasses synthesis and dephosphorylation of 1,3-BPG thus cell is deprived of energy usually obtained here

23

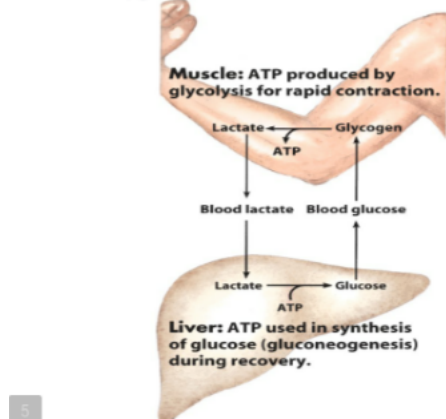
Pyruvate Dehydrogenase and Arsenic Poisoning

- Arsenic can interfere with glycolysis at glyceraldehyde 3-phosphate step, decreasing ATP production
- BUT arsenic poisoning is primarily due to inhibition of enzymes requiring lipoic acid as cofactor, including pyruvate dehydrogenase and α -ketoglutarate dehydrogenase
- Arsenite (trivalent form) forms stable complex with thiol (-SH) group of lipoic acid, making it unavailable as coenzyme
- Pyruvate (and lactate) accumulate
- Particularly affects brain (like pyruvate kinase deficiency) - neurological complications and death

18

17. A question about lactate & oxygen debt....

The Cori cycle "pays the debt" for rapid contraction of muscle, generating lactate:



The Cori cycle

- Lactate (lactic acid) decreases the pH of muscle tissues;
- - Lactate produced causes acidity to develop in muscle tissues, which causes muscle cramps and pain associated with high levels of muscle contraction
- - Lactate must be removed from circulation to maintain efficiency of glycolysis
- - Liver takes lactate produced by anaerobic reduction of pyruvate, and produces glucose by gluconeogenesis (pathway covered in a later lecture)
- - This is called the **Cori cycle**

18. Uses of the TCA cycle..

A. As a center of metabolic pathway for many biomolecules.

B. It's intermediates are used for heme biosynthesis

C. Provides raw materials for glucose biosynthesis

D. Provides raw materials for amino acid biosynthesis

E. All of the above

ANS E

Biological Importance of TCA cycle

- Provides raw materials for glucose biosynthesis (eg. Oxaloacetate)
- Provides precursor for heme biosynthesis (succinylCoA)
- Provides raw material for amino acid biosynthesis (Oxaloacetate used for aspartate and α -ketoglutarate for glutamate synthesis)
- Provides acetylCoA for fatty acid and cholesterol biosynthesis (Citrate is the source of AcetylCoA)
- Useful mainly for energy production from dietary molecules (CHO, fats, amino acids)

12

19. Oxaloacetate can NOT be derived from

- A. Alanine
- B. Arginine
- C. Methionine
- D. Leucine
- E. Serine

Metabolic Classification of Aas		
Ketogenic	Mixed	Glucogenic
Leu	Ile	Ala, Arg, Asp, Asn, Glu, Gln, Ser, Met, Pro, Gly, His, Thr, Val, Cys
Lys	Tyr	
	Trp	
	Phe	

ANS. D,.....Leucine is ketogenic, while the rest of the choices are Glucogenic. Oxaloacetate is derived from Glucogenic amino acids, while compounds containing ketone group such as alpha-Ketoglutarate are derived from ketogenic amino acids.

20. Which one of the ff TCA cycle enzymes is membrane embedded?

- A. Citrate synthase
- B. SuccinylCoA Synthetase
- C. Succinate dehydrogenase
- D. Isocitrate dehydrogenase
- E. Alpha-Ketoglutarate dehydrogenase

ANS. C,.....Succinate dehydrogenase is the only enzyme of citric acid cycle that is embedded in the inner mitochondrial membrane.

21. Which citric acid cycle enzyme is different?

- A. Succinate dehydrogenase
- B. Isocitrate dehydrogenase
- C. Alpha-Ketoglutarate dehydrogenase
- D. Citrate synthase
- E. Pyruvate dehydrogenase

ANS. A,.....Succinate dehydrogenase is different from the rest because it's membrane bounded.

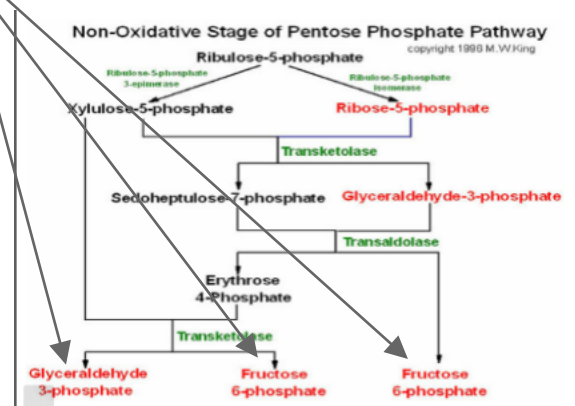
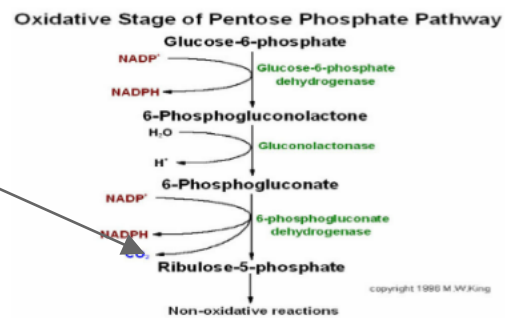
22. True about HMP pathway

- A. It's linked to many synthetic pathways
- B. The oxidative phase produces a 5 carbon compound.
- C. The non oxidative phase provides compounds that can be used for glycolysis
- D. No ATP is directly consumed or produced in the cycle
- E. All of the above.

ANS E

The Pathway...

- It is linked with synthetic pathways & glycolysis
- No ATP is directly consumed or produced in the cycle
- Carbon 1 of Glucose-6-phosphate is removed as CO_2 during the oxidation.



23. Which one of the ff is true about HMP pathway

- A. ATP is needed to drive the pathway
- B. When cells require more energy the oxidative phase converts Ribose-5-P to fructose-6-P and Glyceraldehyde-3-P
- C. When more ribose is required than NADPH the non-oxidative phase runs faster
- D. When more NADPH is required than ribose, the non-oxidative phase runs faster than the oxidative phase
- E. G6PD is a key enzyme that catalyzes the rate limiting step

Regulation of the Pathway

- G6PD is a key enzyme that catalyzes the rate limiting step.
- When more NADPH is required than ribose, the oxidative reaction goes at higher rate than the nonoxidative phase.
- When more ribose is required than NADPH the reverse reaction of the nonoxidative phase and the oxidative phase run faster.
- If cells required more energy the nonoxidative phase converts Ribose-5-P to fructose-6-p and Glyceraldehyde-3-P

ANS. E

24. FALSE about Glycogen

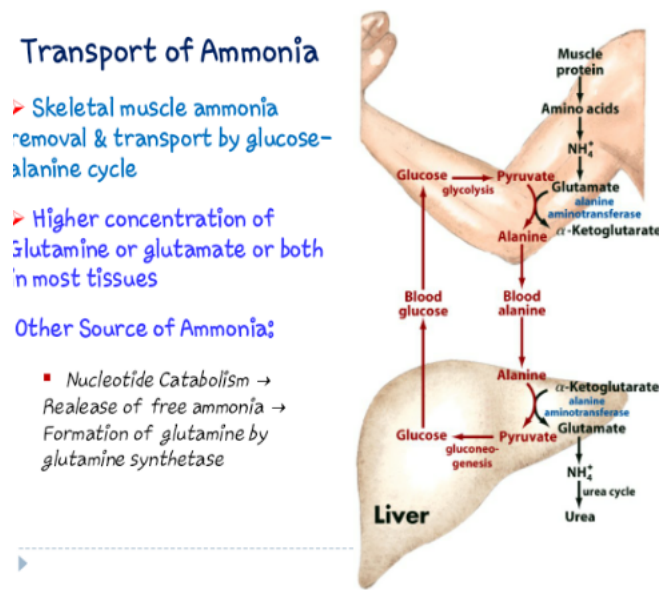
- A. Is a highly branched polymer
- B. Is also called animal starch
- C. Contains only one type of bond.
- E. As an energy source in skeletal muscles

ANS. C,.....Glycogen is a highly branched polymer that contains two types of bonds, alpha -1,4 & alpha -1,6 glycosidic linkages, and also serves as energy storage form in animals.

25. Which of the following conversion takes place in skeletal muscles?

- A. Pyruvate to Alanine
- B. Glutamate to pyruvate
- C. Alanine to pyruvate
- D. Glutamate to Alanine
- E. Alanine to Glutamate

ANS. A,.....In skeletal muscles, Pyruvate is converted to Alanine. Alanine is then transported to the liver in order to be converted back to Pyruvate.



26. Ammonia leaves skeletal muscle and moves into the blood in the form of

- A. Glutamine
- B. Glutamate
- C. Aspartate
- D. Alanine
- E. Asparagine

ANS. D



27. FALSE about Uses of NADPH

- A. For reductive biosynthesis
- B. For reduction of Hydrogen peroxide / H₂O₂/
- C. To generate ATP, by releasing electrons into the Electron transport chain.
- D. Used during phagocytosis
- E. In synthesis of NO

ANS. C IS FALSE, It's NADH and FADH₂ that release electrons into the Electron Transport Chain/ETC/ for production of ATP

Uses of NADPH...

- A. **Reductive Biosynthesis**: the electrons in this coenzyme are used during biosynthesis of cholesterol, fatty acids etc
- B. **Reduction of H₂O₂**: this reactive oxygen species (ROS) is produced in cells but when found in excess can damage cells. It damages DNA, proteins, unsaturated lipids and leads to cell death. NADPH is required for the reduction of oxidized glutathione by glutathione reductase.

Use NADPH...

- C. **Cytochrome P₄₅₀ system**: NADPH is useful in the action of monooxygenases during reduction and oxidation processes

$$R-H + O_2 + NADPH + H^+ \rightarrow ROH + H_2O + NADP$$
- D. **Phagocytosis by white blood cells**: this process also requires NADPH especially the reaction by NADPH oxidase that converts oxygen to superoxide. The superoxide in turn changes to water by superoxide dismutase.
- E. **Synthesis of NO**: is a molecule that is biologically useful as vasodilator but if in excess causes oxidative damage. NO synthase requires NADPH for synthesis.

28. Cori cycle is used to

- A. Synthesize Glucose through gluconeogenesis using lactate as a substrate
- B.....
- C.....
- D.....
- E.....

ANS. A

The Cori cycle...

The Cori Cycle

- One fate of pyruvate produced in skeletal muscle is its conversion to lactate and to glucose in liver.
- Strenuous muscle undergoes anaerobic glycolysis in addition to aerobic one.
- This produces large quantity of L-lactate which causes muscle fatigue.
- Lactate is cleared from skeletal muscle transported to liver via blood.
- L-lactate is converted enzymatically to glucose through the process of gluconeogenesis.
- This glucose is transported to skeletal muscle via blood for further energy production.
- This cyclic process is termed the Cori cycle.

29. Glucagon stimulates which ANABOLIC reaction?

- A. Glycolysis
B. Glycogenesis
C. PPP/Pentose phosphate pathway/
D. Gluconeogenesis

ANS D,.....Glucagon stimulates the formation of Glucose, by favoring the occurrence of Glucose producing processes such as: glycogenolysis, gluconeogenesis, lipolysis and by disfavoring processes that decrease the amount of glucose, such as glycolysis, glycogenesis and lipogenesis.

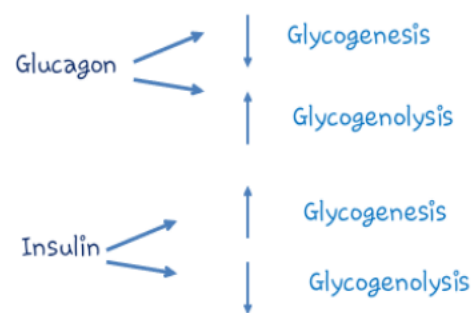
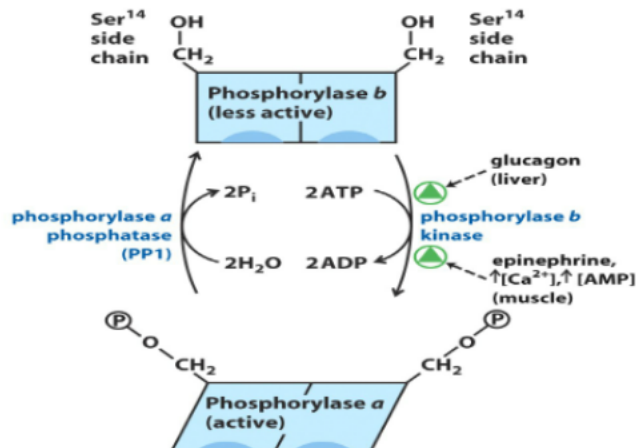
Effect of glucagon on glycogen metabolism

- Glucagon is released during starvation and through a signaling pathway phosphorylates cytosolic enzymes including glycogen synthase and glycogen phosphorylase.
-
- Phosphorylation makes glycogen synthase inactive but activates glycogen phosphorylase.

Effect of Insulin on glycogen metabolism

- Insulin is released in response to well-fed state.
- Insulin works via activation of protein phosphatases and phosphodiesterase.
- The former dephosphorylate phosphorylated proteins but the latter convert cAMP to AMP.
- When both glycogen phosphorylase and glycogen synthase are dephosphorylated the former becomes inactive but the latter becomes active

Glycogen phosphorylase is regulated by phosphorylation on Serine residues:



Note that insulin and glucagon regulate glycogen metabolism reciprocally by reciprocally activating the two key enzymes

Very Important!!!

30. Deficiency of glucose 6 phosphatase causes

A. Von Gierke disease

B. Pompe disease

C. McArdle

D..... E.....

ANS. A,.....Von Gierke disease type Ia, is caused by deficiency of glucose 6 phosphatase

McArdle Syndrome

- Due to deficiency of skeletal muscle glycogen phosphorylase.
- Skeletal muscle is affected liver is normal.
- Temporary weakness and cramp of skeletal muscle.
- No rise in lactate during exercise but high level of glycogen in skeletal muscle.
- High myoglobinemia and myoglobinuria.

Von Gierke disease

- Has two types: one due to glucose 6-phosphatase deficiency (Type Ia) and another due to glucose 6-phosphate translocase deficiency (Type Ib)
- This results in increased glycogen storage in liver, kidney and intestine.
- Clinical manifestations include: severe fasting hypoglycemia, fatty or enlarged liver, renal disease, growth retardation and delayed puberty.

Pompe disease

- Lysosomal storage disease caused by deficiency of $\alpha(1 \rightarrow 6)$ Glucosidase deficiency.
- An inborn defect that affects liver, heart and muscle cells (excessive glycogen in vacuoles).
- Normal glycogen structure and normal glucose level.
- Clinical manifestations: massive cardiomegaly, early death due to heart failure.

31. Uronic acid pathway is used in

A. Conjugation to less polar compounds

B. Synthesis of GAGs(glycosaminoglycans)

C. Occurs in cytosol

D. Enables Xylulose to enter the central pathway

E. All of the above

ANS. E

Uronic acid pathway

- It is an alternate pathway of glucose metabolism
- It converts glucose to glucuronic acid and ascorbic acid
- Occurs mainly in the liver cytosol
- Not involved in energy production

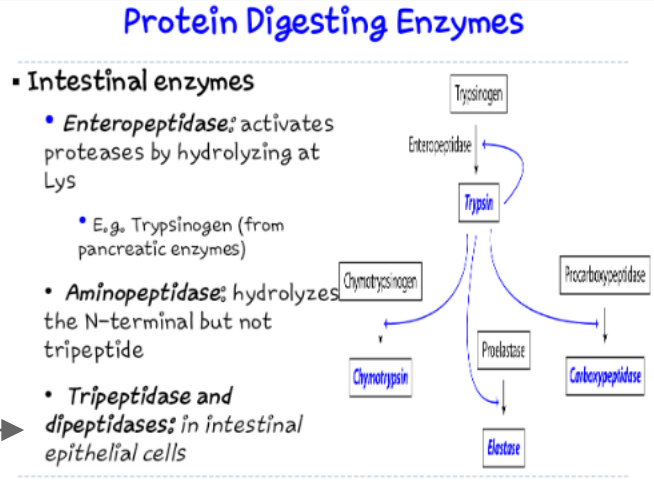
Biological importance

- ✓ **Conjugation to less polar compounds** as bilirubin, steroids and some drugs making them more water soluble (detoxicated).
- ✓ **Synthesis of glycosaminoglycans** (mucopolysaccharide) **and the extracellular matrix** as heparin, hyaluronic acid.
- ✓ In plants and some animals (not Human) glucuronic acid serves as a **precursor of L-ascorbic acid**.
- ✓ The uronic acid pathway also provides a mechanism by which dietary **D-xylulose** enter the central pathway.

32. Which of the ff protein degrading enzyme belongs to dueodenal secretions?

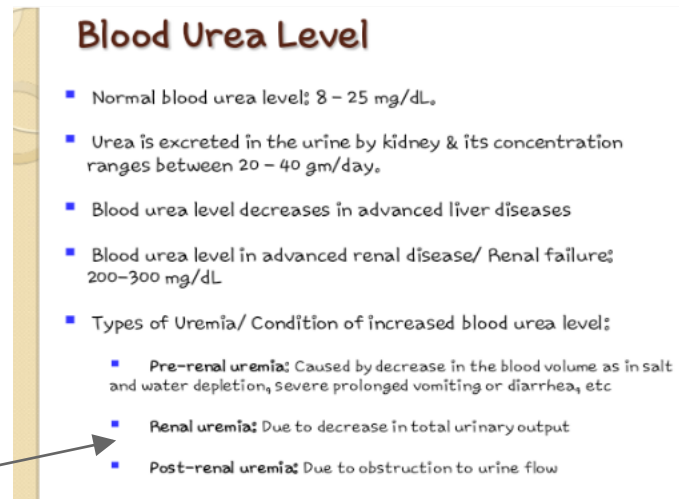
- A. Chemotrypsin
- B. Elastase
- C. Trypsin
- D. Carboxypeptidase
- E. Dipeptidase

ANS. E



33. Renal uremia occurs during

- A. Decrease blood volume
- B. Prolonged vomiting
- C. Kidney failure
- D. Obstruction of Urine flow
- E.....



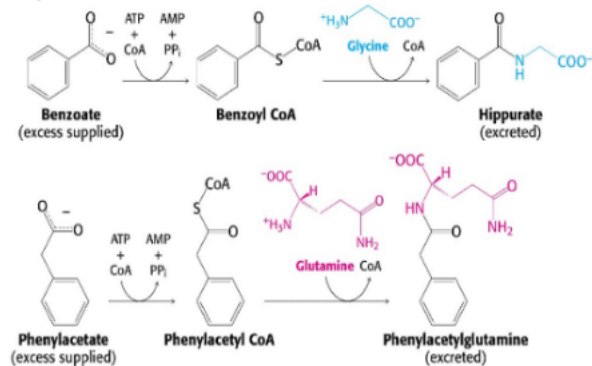
ANS. C,.....Renal uremia occurs as a result of decrease in total urinary output. Such a condition occurs due to Renal failure

34. Benzoate is given to patients with defective urea cycle in order to remove ammonia in the form of.

- A.
- B. Hippuric acid
- C. Phenylacetylglutamine
- D.
- E.

Possible treatments for defective urea cycle

Feeding the patients with Benzoate or phenylacetate

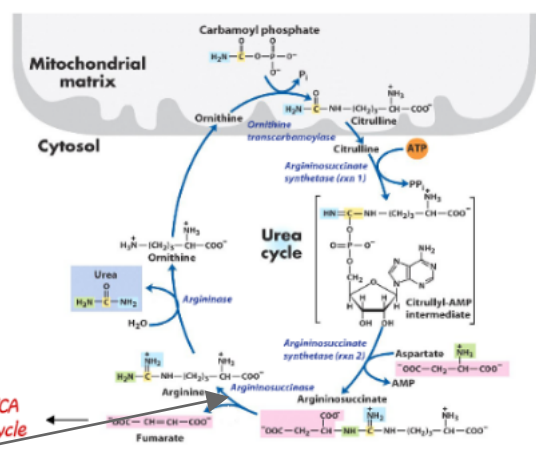


ANS. B,.....feeding Benzoate in excess amounts to patients with defective urea cycle results in the removal of ammonia through formation of Hippurate, while feeding them Phenylacetate results in the formation of phenylacetylglutamine, another compound helpful for the removal of ammonia.

35. A patient is advised to ingest high amounts of L-arginine because of the most likely lack of which enzyme?

- A. Argininosuccinate synthetase
- B. Argininosuccinase
- C. Carbomoyl phosphate synthetase
- D. ...
- E...

UREA CYCLE: Formation of ammonia

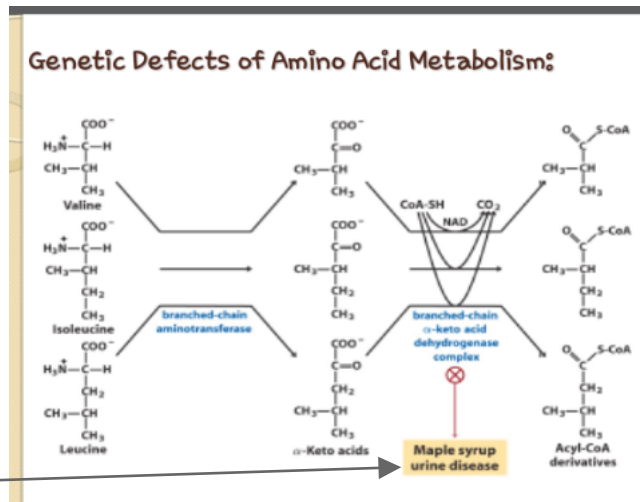


ANS. B,.....deficiency in Argininosuccinase enzyme results in decreased production of Arginine. Patients with this condition are advised to ingest L-arginine in high amounts so as to compensate the loss.

36. Maple syrup urine disease results from defect of which enzyme

- A. Branched chain aminotransferase
 B. Branched chain Alpha-keto acid dehydrogenase complex.
 C.....
 D.....
 E.....

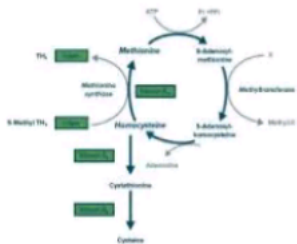
ANS. B



37. Homocystinuria is caused by deficiency of cystathionine synthase in which of the ff cycles.

- A. Methionine metabolism cycle B..... C..... D..... E.....

ANS. A



Homocystinuria is a disorder of methionine metabolism, leading to an abnormal accumulation of homocysteine and its metabolites (homocystine, homocysteine-cysteine complex, and others) in blood and urine.

Homocystinuria

- Deficiency of **cystathionine synthase**, the pathway for cysteine synthesis
- High urinary levels of homocysteine, a substrate of the impaired enzyme.
- High plasma homocysteine causes chronic progressive skeletal abnormalities, mental retardation and severe thrombotic tendencies.
- **Treatment by two forms**
 - by high doses of vitamin B6
 - limiting intake of methionine

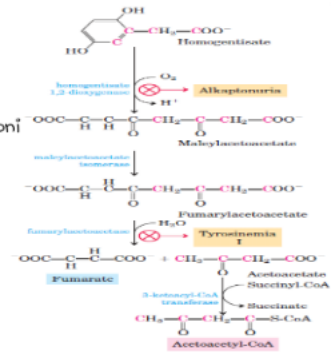
38. Tyrosinaemia type 1 is caused by lack of

- A. Fumerylacetoacetate
- B. Homogenitase-1,2-dioxygenase
- C. Tyrosine aminotransferase
- D. Phenylalanine hydroxylase
- E. P-Hydroxyphenylpyruvate dehydrogenase

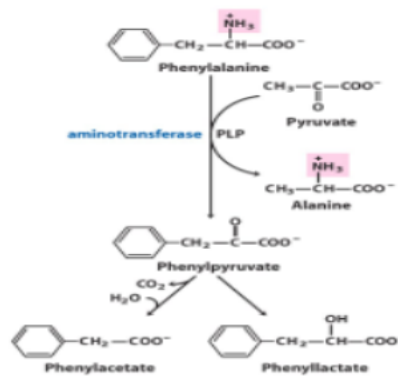
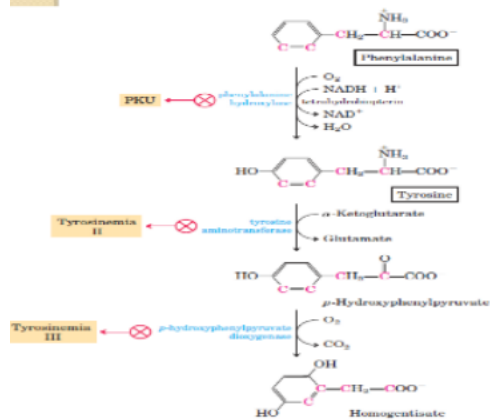
ANS. A

Tyrosinaemia type 1

- Usually present with severe liver disease and renal Fanconi syndrome.
- Dark urine when stay



Phenylketonuria (PKU) and Tyrosinemia



- PKU: Can cause mental retardation (major manifestation).
- Treatment: low phenylalanine diet.
- Tyrosinemias - if untreated, weakness, liver damage, mental retardation.

* deficiency of Phenylalanine hydroxylase results in Phenylketouria/PKU/

* deficiency of Fumerylacetoacetate results in Tyrosinemia I

* deficiency of Tyrosine aminotransferase results in Tyrosinemia II

* deficiency of P-Hydroxyphenylpyruvate dehydrogenase results in Tyrosinemia III

39. Deamination takes place in

A. Liver. B. Skeletal muscle C. Adipose tissue D. Brain E....

ANS. A,.....In the human body, deamination takes place primarily in the liver, however glutamate is also deaminated in the kidneys

40. Removal of ammonia from side chains(R-groups) of compounds is called

A. Deamidation

B. Transamination

C. Deamination

D.....

E.....

Other Catabolic Pathways ... (Cont'd)

- **Amidation/ Deamidation:**
 - Glutamine & asparagine contain R group amides
 - Released as NH_4^+ by deamidation
 - Asparagine is deamidated by asparaginase,
 - Yielding aspartate and NH_4^+
 - Glutaminase acts on glutamine, important in kidney: Forming glutamate & NH_4^+
- **Hydrolytic deamination:** important for deamination of nitrogenous bases
- **Reductive deamination:** by addition of 2H atoms to form corresponding carboxylic acid (such as fatty acid)

ANS. A,.....Deamidation is a chemical reaction in which an amide functional group in the side chain of the amino acids asparagine or glutamine is removed or converted to another functional group.

41. FALSE about alpha-Keto glutarate

A. Can be used for gluconeogenesis

C. Is an intermediate of TCA cycle

B. Donor of ammonia

D.....

E.....

ANS. B,.....It's an acceptor of ammonia, NOT donor.

42. Cimetidine blocks gastric acid secretion by inhibiting

A. Histidine

B. Histamine

ANS. B, Histamine

Biosynthesis of Histamine

Synthesized and released by mast cells
Mediator of allergic response: (H1 receptors)

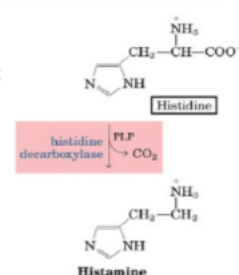
H1 blockers:

Diphenhydramine (Benadryl)
Loratidine (Claritin)

stimulates secretion of gastric acid (H2 receptors)

H2 blockers: Cimetidine (Tagamet)

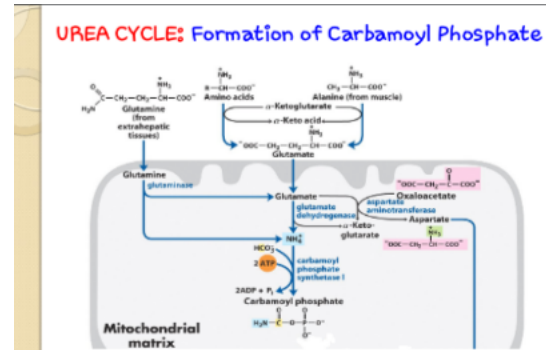
Ranitidine (Zantac)



43. Which of the ff reaction of urea cycle takes place in the mitochondria?

- A. Synthesis of carbomoyl phosphate
 B. Synthesis of ornithine
 C..... D..... E.....

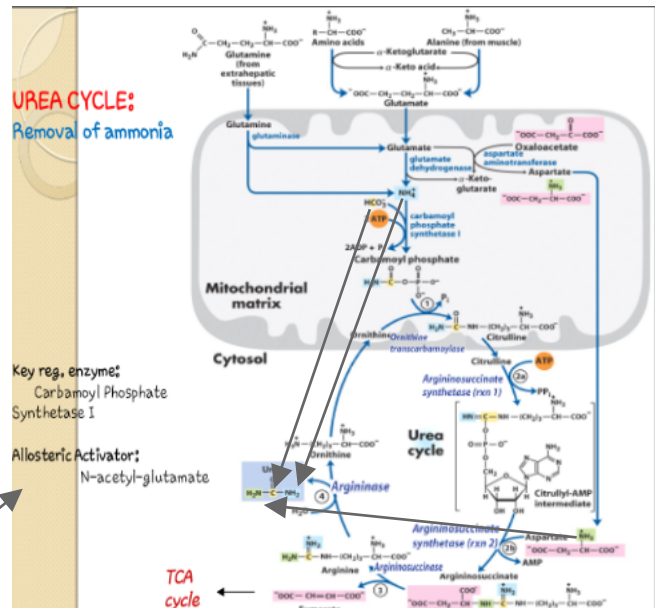
ANS. A



44. The carbon in urea is derived from

- A. CO₂(HCO₃⁻)
 B.... C..... D..... E...

ANS. A



45. The two nitrogens in urea are derived from

- A. Aspartate and ammonia
 B..... C..... D..... E.....

ANS. A

46. Which one of the ff stimulates carbamoyl synthetase I during urea cycle.?

- A. N-acetyl-glutamate B..... C..... D..... E.....

ANS. A

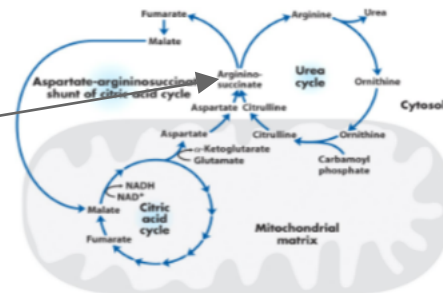
47. Which compound links urea cycle and citric acid(TCA) cycle?

A. Argininosuccinate

B..... C..... D..... E.....

ANS. A

Linkage of Citric Acid and Urea cycles/
Fumarate (in cytosol) → Citric acid cycle (in mitochondria)



48. Which subunit of the ATP synthase does Oligomycin inhibit?

A. F₀

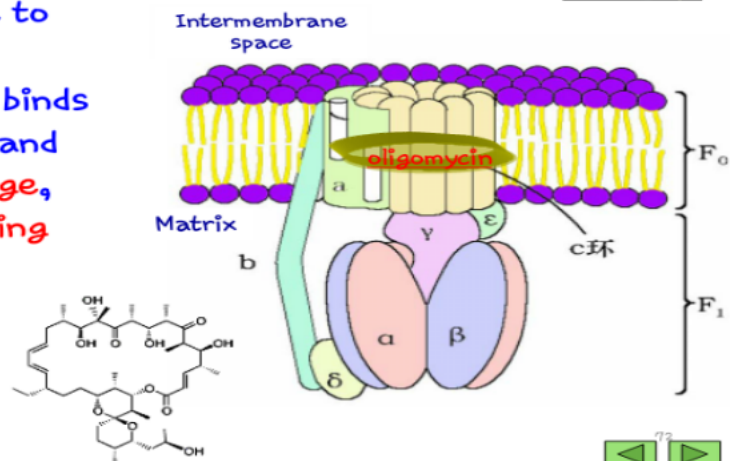
B. F₁

C. B₂

ANS. A

Oligomycin: Inhibitors of ATP Synthase

• F₀ is sensitive to oligomycin, an antibiotic that binds in the channel and blocks H⁺ passage, thereby inhibiting ATP synthesis



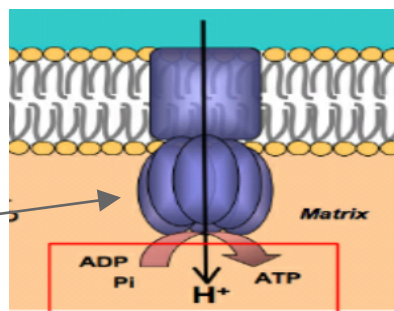
49. Which subunit of ATP Synthase uses ADP to make ATP?

A. F₀

B. F₁

C. B₂

ANS. B



50. Which one of the ff doesn't happen during longterm physical exercise

- A. Red blood cells number increases
- B. Surface area of lung increases
- C. Blood flow towards the lung increases
- D. Oxygen diffusion capacity of lungs increases
- E.....capillaries

ANS. A

51. Clinically, A man is unable to use TAG and there is an accumulation of TAG in his muscle. As a result muscle weakness. What is the possible deficiency?

- A. Carnitine B. Lipoprotein lipase C..... D..... E.....

ANS. A(Not sure!)

52. Pancreatic lipase is activated by

- A. Colipase/ Ca^{++} B. Bile salt C..... D.... E.....

ANS. A

53. Which one of the ff is NOT correctly paired

- A. TAG - Phospholipase C. PL(Phospholipid) - Pancreatic lipase
- B. Cholesterol - Esterase D. PL(Phospholipid) - Phospholipase A2

ANS. Not sure!

54. What is the reason of ketosis in uncontrolled DM(Diabetes Mellitus) type 1

- A. Inhibition of FA synthesis
- B. Decrease in OAA(Oxaloacetate)
- C. Increase in HMG CoA synthase
- D. Increase in HMG CoA reductase
- E. All of the above

ANS. B(Not sure!)

55. Which is FALSE about Acetone?

- A. It is released in expired air
- B. Tissues can't use it as a fuel for energy
- C. It is important for the diagnosis of DM
- D. It cant produce acidosis
- E. It is used as a fuel for CNS

ANS. E

56. Which one of the ff tissues is INCORRECTLY paired with its source of energy?

- | | |
|-------------------------|--------------------------|
| A. Liver - FA | D. Brain - ketone bodies |
| B. Muscle - FA | E. All |
| C. RBCs - Ketone bodies | |

ANS. C,.....RBCs and Liver cannot use Ketone bodies, but Liver is the main site of ketone body synthesis.

57. In an increased fatty acid synthesis, which one would probably decrease beta oxidation of FA

- A. Decrease in adipocyte lipolysis
- B. Inhibiting the translocation of cellular components
- C.....
- D.....
- E.....

ANS. B

58. Which one is NOT correct about ATP synthesis in oxidative phosphorylation?

- A. It occurs in a series of electron carrier
- B. It binds molecular oxygen as a complex
- C. It occurs in the matrix side of mitochondria
- D. It synthesizes ATP
- E. It has ATPase activity

ANS. B

59. Rate limiting enzyme of lipogenesis/ fatty acid synthesis

ANS. Acetyl CoA carboxylase

60. Absence of apo B-100 found in LDL receptor results in/ LDL receptor is defective in/

ANS. Familial hypercholesterolemia

61. Defect in the formation of chylomicrons is associated with

ANS. Abetalipoproteinemia,.....The lack of triglyceride transfer activity leads to the disease abetalipoproteinemia. This disease affects both chylomicron assembly in the intestine and VLDL assembly in the liver.

62. The function of fatty acid binding protein(FABP)

ANS. Transport fatty acids

63. How many ATP can be produced from oxidation of one beta-hydroxy butyrate?

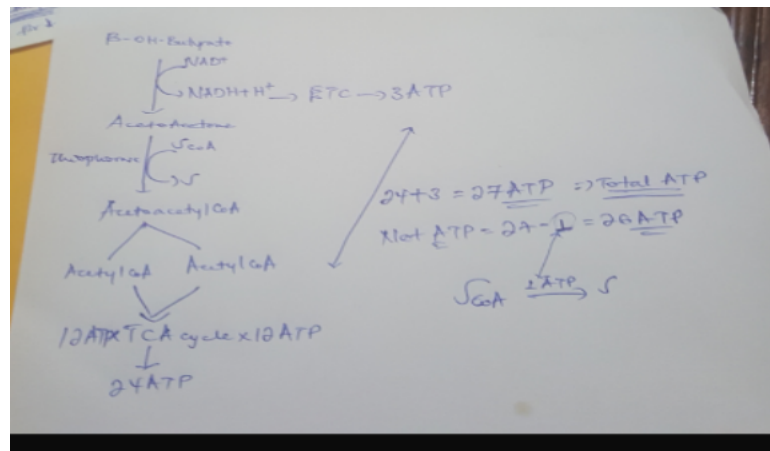
A. 22

B. 23

C. 27

D. 25

ANS. C



64. Presence of fat in stomach causes the release of all of the following EXCEPT

A. CCK

B. Secretin

C. Protease

D. GIP

E. GLP-1

ANS. C

65. Conjugation of bile salt involves

A. Glycine and Glutamine

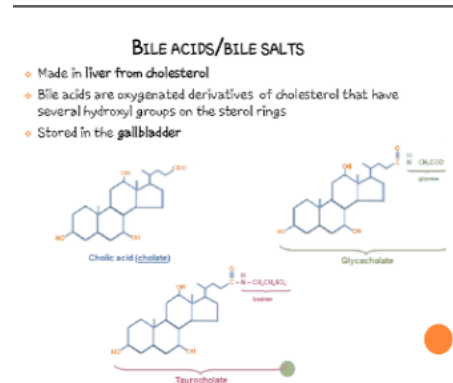
B. Taurine and Glycine

C.....

D.....

E.....

ANS. B,.....Glycine, Cholat and tautomeric, these three are used in the conjugation of bile acids.



66. What do uncouplers do?

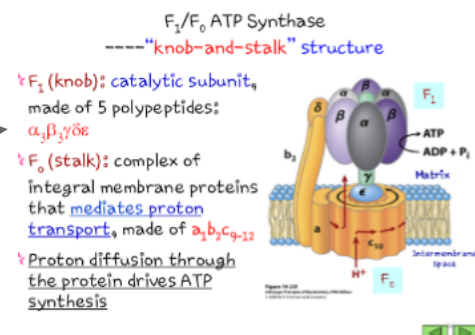
ANS. They let Electron transport chain/ETC/ continue without ATP synthesis

Uncouplers

- Uncouplers** disrupt the tight coupling between electron transport and the **ATP synthase**.
- They act by **dissipating the proton gradient** across the inner mitochondrial membrane created by the electron transport system, so **affect ATP synthesis**.

67. The Catalytic subunit of ATP synthase F₁

ANS.



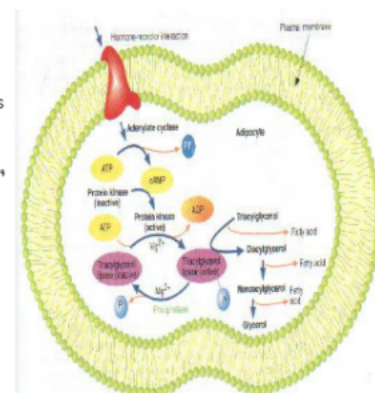
68. What does Hormone sensitive lipase /HSL/ do?

ANS. The main function of hormone-sensitive lipase is to mobilize the stored fats. HSL functions to hydrolyze either a fatty acid from a triacylglycerol molecule, freeing a fatty acid and diglyceride, or a fatty acid from a diacylglycerol molecule, freeing a fatty acid and monoglyceride. HSL is activated as a result of phosphorylation stimulated by the secretion of Glucagon or epinephrine.

MOBILIZATION OF STORED FATS: LIPOLYSIS

□ Non-esterified fatty acid release from the adipocytes is initiated by the action of hormone sensitive lipase (HSL), which begins to hydrolyze the stored triglyceride.

□ The final products of triacylglycerol hydrolysis are glycerol and nonesterified fatty acids.



69. There was a question about reasons for ketosis.

Which one of the ff is NOT a cause for ketosis?

ANS. High carbohydrate diet.

***It's Low carbohydrate diet that can be a cause for ketosis**

know every possible reason

CAUSES OF KETOSIS

- ☐ Uncontrolled diabetes mellitus (Type I)
- ☐ Starvation
- ☐ Chronic alcoholism
- ☐ Von-Gierke's disease
- ☐ Heavy exercise
- ☐ Low carbohydrate diet- For weight loss
- ☐ Pyruvate carboxylase deficiency

70. What facilitates B-oxidation?

Read about the functions of oxidation of fatty acids.

***Fatty Acid Oxidation Provides Energy for Cellular and Metabolic Work**

***Fatty Acid Oxidation Provides Fuel to the Brain during Starvation**

***Fatty Acid Oxidation Generates Heat**

***Fatty Acid Oxidation Generates Intermediates in Synthetic processes**

71. Questions about carnitine deficiency, familial hypercholesterolemia, abetalipoproteinemia (these were either questions or answers. The questions were presented in case study type of way.)

Chylomicrons, HDL or LDL? Study the TAG and cholesterol contents of these three; which one has the more, which one has less?

72. The energy produced from the removal of Pi from ATP

ANS. -30.5 KJ/mol or -7.3 KCal/mol

73. How does propionyl CoA gets enter into gluconeogenesis?

ANS. propionyl CoA, a precursor for succinyl CoA, can be converted to pyruvate and enter into gluconeogenesis.

Conversion of propionyl CoA to succinyl CoA, Succinyl CoA, an intermediate of the TCA cycle, can form malate, which can be converted to glucose in the liver through the process of gluconeogenesis. Certain amino acids also form glucose by this route

74. Carnitine is synthesized in Liver and kidney from amino acids

ANS. Lysine and Methionine

75. Under standard condition

Which is true?

A. free energy..... =1

B. change in G.....= 0

C..... D..... E.....

It asked what will be the result of G/G_0 under standard condition

76. Which of the ff is TRUE?

ANS. Liver CAN'T utilize ketone bodies because it lacks the enzyme THIOPHORASE. Thank the Lord! Because liver happens to be the major site of ketone body synthesis. If liver had Thiophorase, futile reactions may happen.

*LIVER LACKS THIOPHORASE BUT HAS GLYCEROL KINASE

*ADIPOCYTES LACK GLYCEROL KINASE BUT HAVE THIOPHORASE

77. What affects HMG-CoA reductase used in Cholesterol Biosynthesis

ANS. Statins,.....Statins inhibits HMG-CoA reductase

78. Colipase increases the activity of pancreatic lipase by

- A. Binding to pancreatic lipase to avoid other inhibitor substances
- B. Forming pancreatic lipase-colipase complex.
- C. By avoiding binding of bile salts

ANS. A or C (Not sure)

79. A question about problem of Medium Chain FA transport to mitochondrial membranes?

TYPES OF FATTY ACYL CoA DEHYDROGENASES

- ☐ Long chain fatty acyl CoA dehydrogenase (LCAD) acts on chains greater than C12.
- ☐ Medium chain fatty acyl CoA dehydrogenase (MCAD) acts on chains of C6 to C12.
- ☐ Short chain fatty acyl CoA dehydrogenase (SCAD) acts on chains of C4 to C6.

↓
MCAD deficiency is thought to be one of the most common inborn errors of metabolism.

80. A question calculation yalebet, as an example ye adenine amount be percent yesetina then calculate ye guanine'n or cytosine'n or thymine'n

81. Who proposed a double helical model of DNA?

ANS. James Watson & Francis Crick, Watson-Crick model of DNA.

82. The helical structure of DNA repeats after how many residues on each chain?

ANS. 10

83. What happens to double stranded DNA when temperature is raised?

#know these, these things happen when temperature is raised

- *Local unwinding of the ds DNA.

- *This occurs preferentially in regions where the two strands are held together less strongly.

- *In these regions the strands separate to form bubbles of ss regions.

- *The DNA sequence in these regions is enriched in A/T base pairs because the interactions between the two strands are weaker in A/T rich regions.

- *In G/C rich regions strands are held together more strongly so they don't unwind until higher temperatures.

84. Which of the ff is true about DNA replication?

- A. Both the leading and lagging strands copy continuously
- B. Both the leading and lagging strands copy in segments
- C. The leading strand copies in segments while the lagging strand copies continuously
- D. The leading strand copies continuously while the lagging strand copies in segments

ANS. D

85. Which of the ff enzymes removes RNA primers and adds DNA nucleotides into a growing strand of DNA

ANS. DNA Polymerase III

86. Which one of the ff is a stop/termination condon?

ANS. Stop condons - UGA, UAG & UAA

Start condon - AUG, codes for methionine

87. How many codons can code for specific aminoacids?

- A. 45
- B. 48
- C. 64
- D. 61
- E...

ANS. D

88. Degeneracy of the genetic code states

ANS. A given amino acid must have more than one triplet coding for it.

89. The Shine-Dalgarno sequence of mRNA is

UANS. Complementary to the sequence at the 3' end of 16S rRNA

90. An activity intrinsic to the 23S rRNA found in the 50S ribosomal subunit

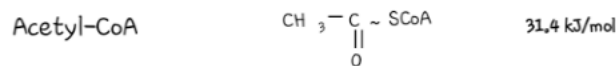
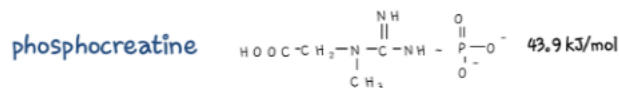
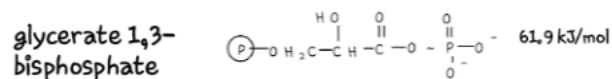
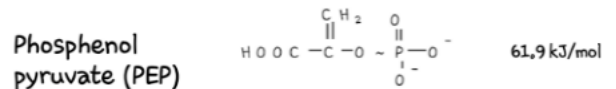
ANS. Peptidyltransferase activity,..... The larger subunit(50S) of prokaryotes has a peptidyltransferase activity.

91. A question about one of the three types of mutations; Silent mutation, Missense mutation or Nonsense mutation

92. Which of the ff is a high energy phosphate compound?

ANS. PEP2 or 1,3-bisphosphoglycerate

High-energy Compounds



93. Which of the ff is involved in a single carbon transfer?

A. Lipoic acid B. Tetrahydrofolate C. CoA D..... E.....

ANS. B

94. Which coenzyme is derived from pantothenic acid (vitamin B5)

ANS. Coenzyme A / CoA/

95. A 50 years old man with prostate cancer minamin.....should be diagnosed with an elevated plasma level of

ANS. Acid phosphatase

96. Groups that are tightly bound to an enzyme to increase its activity are called

ANS. Prosthetic groups

97. A coenzyme for enzymes that transfers Carboxy groups or the most oxidized for of carbon / CO₂/ is

A. CoA

C. Pyridoxal Phosphate/PLP/

B. Biotin

D. Thiamine pyrophosphate /TPP/

ANS. B

98. Which of the ff can be used as a store of energy by the brain

ANS. Creatine Phosphate,...Creatine phosphate is an important energy store in skeletal muscle and in the brain.

99. How many protons are moved out of the matrix into the intermembrane space when electrons pass through Complex IV

- A. 4 B. 6 C. 3 D. 5 E. 2

ANS. E,.....4 protons from Complex I, NO protons from Complex II, 4 protons from complex III and 2 protons from Complex IV.

100. Cyanide inhibits which complex of the ETC?

- A. I B. II C. III D. IV

ANS. D

101. Which one of the ff is NOT activated during glycogen degradation?

ANS. Insulin

102. The committed step in de novo Pyrimidine synthesis is

ANS. The Carbamoyl phosphate synthetase-II /CPS-II/ catalyzed reaction

103. A drug that is given for a patient with heart attack is

ANS. TPA /Tissue plasminogen activator/

104. A question that asks about the relationship between change in G and K_{eq}

Know this * Change in $G > 0$ then, $K_{eq} < 1$

* Change in $G < 0$ then, $K_{eq} > 1$

* Change in $G = 0$ then, $K_{eq} = 1$

105. A competitive inhibitor that resembles Succinate

ANS. Malonate

106. A question about raw materials for gluconeogenesis

***The Raw materials include:**

Pyruvate

Glycerol

L-Lactate

L-Alanine

L- Glutamate

Intermediates of TCA cycle

107. The TATA box region on the DNA is a binding site for

A. DNA dependent RNA polymerase

B. DNA dependent DNA polymerase

C. RNA dependent DNA polymerase

D. RNA dependent RNA polymerase

E.....

ANS. A

108. Rotenone inhibits complex I by inhibiting transfer of electrons from

- A. Fe-s to cyt c
- B. Fe-s to ubiquinone
- C. cyt c to ubiquinone
- D. Cyt c to Fe-S
- E.....

ANS. B

109. Long distance runners are advised to eat

- A. Legumes and protein rich foods in order to build muscles
- B. CHO (...and pasta)rich foods glycogen'n lemetekat
- C. Vitamins
- D..... E.....

ANS. B

For most, the 60/25/15 diet is a good rule **of** thumb. This **of** course means that carbohydrates should form the cornerstone **of** your diet. Since carbo's are the most important energy source for **long-distance running**, it's probably no surprise that so many **runners eat** hefty portions **of** pasta, rice, bread and potatoes. Mar 23, 2002



www.coolrunning.com » engine

[The Runner's Diet - Cool Running](#)

110. Oxygen

- A. Has high negative redox potential, so it easily reduces other molecules
- B. Has high positive redox potential, so easily reduces other molecules
- C. Has high positive redox potential, so easily oxidizes other molecules
- D. Has high negative redox potential, so easily oxidizes other molecules

ANS. C

111. On the beta oxidation of stearic acid (18:0), how many of total ATPs can be formed.

- A. 148 B. 102 C..... D..... E.....

ANS. A

112. The rate limiting enzyme during ketone bodies utilization is

- A. Thioestrerase B. Thiolase C. Thiophorase D... E...

ANS. C

113. Which organ cannot utilize ketone bodies

- A. Kidney B. Brain C. Liver D. Adipose tissue E. Skeletal muscle

ANS. C,.....because liver lacks thiophorase. RBCs can't do that also because they lack mitochondria.

114. Which is FALSE about hexokinase and glucokinase?

- A. They catalyze the same reaction.
B. Glucokinase is found only in the liver
C. Hexokinase has higher K_m for glucose than glucokinase
D.....
E.....

ANS C,....Hexokinase has lower K_m and HIGHER affinity for glucose than glucokinase

115. Which of the following signals for termination during transcription?

- A. Sigma factor B. Rho factor C..... D..... E.....

ANS. B

116. Which one of the ff is involved in reverse cholesterol transport?

- A. HDL B. LDL C. Chylomicrons D..... E.....

ANS. A

117. Which of the ff takes place in the mitochondria during the urea cycle?

- A. Carbomoyl synthesis
B. Argininosuccinate synthesis

ANS. A

118. Total absence of the hypoxanthine-guanine phosphoribosyl transferase enzyme results in a condition

ANS. Lesch Nyhan syndrome

119. Which of the following is NOT required for fatty acid synthesis?

- A. NADPH B. BIOTIN C. COBALAMIN D. CO₂ E...

ANS C

120. Which of the following fatty acid synthesis enzyme regulates substrate availability for Beta-oxidation mechanism

- A. AcylCoA dehydrogenase B. AcylCoA carboxylase.

ANS. B

121. The size of tRNA...

- A..... B.....
C. 50-100 nucleotides long
D..... E.....

ANS. C

122. Which one of the following is a down regulation method for number of LDL receptors?

- A. Intracellular accumulation of LDL
B. Extracellular accumulation of LDL
C..... D..... E.....

ANS. A

123. Which of the following tissue lacks glycerol kinase (glycerol phosphate kinase)?

- A. Liver B. Adipose C..... D..... E.....

ANS. B

124. The source of glycerol 3 phosphate in the Adipose tissue is

- A. Glycerol
- B. Dihydroxyl acetone phosphate from glycolysis
- C D..... E.....

ANS B,.....In the liver bihon it's A because LiVeR has Glycerol kinase

125. Which of the following is true about F0 meselegn and oxidative phosphorylation?

For test two, the answer is it has atpase activity and the componenets are located on the matrix side



LAB QUESTIONS

1. Which of the following is not used to determine total cholesterol?

- A. Cholesterol oxidase
- B. Cholesterol esterase
- C. HMG CoA reductase
- D. Peroxidase
- E. Hydrogen peroxide

ANS. C

2. Which one of the following is one of the most sensitive tests to diagnose renal function?

- A. Urea test B. Creatinine C. Creatine D. Creatinine Clearance E...

Not sure lelochu choices gn the specific answer is CREATININE CLEARANCE test

3. Cholesterol level is said to be high in the human serum sample is it's greater than

- A. 220 B..... C..... D. 239 E. >239

This question is not correct 2 mels new yalew, yetarenew be E neber

4. Which one of the ff test is used in the management of diabetes as a biomarker of long term glycemic control?

- A. Glycated-hemoglobin(Ghb)
- B. Fasting plasma glucose test
- C. Oral glucose tolerance test
- D..... E.....

ANS. A

5. A question asking about the relationship between enzyme AST/ALT and something

- A. Directly
- B. Inversely
- C. Constant
- D...

ANS. B

"CORRECTION" QUESTIONS

9 corrections

*Kenezi wust 3'tu yewetalu ke final wust(119-3=116)

1. Which one of the ff doesn't take place in the post transcriptional modification of RNA

A. Introns removal

B. Splicing

C...

D. BASE MODIFICATION

E

Be D neber yetarenew, but hulum meles slemihonu yeserezal tyakew

2 & 3 ..yezana gize 2 tyake lik adelum yalew yeserezalu

*Yekerut 6 tyakewoch demo

1. All of the ff organs lay transamination yekahedal during catabolism of minamin.....Except

A. Liver

B. Brain

C. Skeletal muscle

D. Adipose tissue

E.kidney

-Yehe tyake be A(LIVER) neber yetaremerw

- Be D enditarem correction yederegal gn Mr. SISAY kayew behuala

2. Cholesterol level is said to be high in the human serum sample if it's greater than

A. 220

B. ..

C...

D. 239

E. >239

This yetarenew be E new

*Be D and E enditarem tesmantenal(Lab sewoch kayut behuala yewesenal)

Ena yehen tyake from other references or videos aytachuh lela meles yemelesachuh temariwoch, ke andu lab lela lab slemileyay memeles yeneberegachuh according to our own lab neber

3. All of the ff are true about nonsense codons except

A.

B. They act as chain termination signals

C..... D.....

E. They can code for amino acids

-yehe tarmo yeneberew be B new.

* Be D enditarem yederegal

4. CCAAT box is minamin true mnarmn

A. ...

B. Prokaryotic lay 35 bp upstream new migegnew

C...

D. Eukaryotic promoter...70-80bp upstream new migegnew

E..

-this be B neber yetaremew

* be D enditarem yederegal

5. Which one of the following minamin is used in TMP de novo synthesis?

A. N10-formyltetrahydrofolate.

B....

C. N5,N10-Methylenetetrahydrofolate

D...

E..

-Be A neber yetaremew

* Be C enditarem yederegal

6. Which of the ff coenzyme is needed for the reduction of ribonucleotides(NDP) to deoxyribonucleotides(dNDP)

A. FMN

B. FAD

C. NAD

D. NADPH

E. TPP

- This be D neber yetaremew

* Be B and D Enditarem yederegal

GOOD LUCK!!!

